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Optimization Models with Probabilistic Constraints

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- Probabilistic constraints model
- MILP reformulation with discrete distribution
- Convex reformulations with special structures
- Monte Carlo sampling approach
- Exercises



- **No recourse actions** is possible in some cases
 - irreversible investment decisions
 - lost lives caused by wrongly planned dams
- The feasibility of here-and-now decisions plays very important role (even more than optimality...)
- Consider points that are feasible with only some probability level

models with probabilistic constraints (chance-constraints)



- Assume no recourse, i.e., shortage cannot be covered by immediate market purchases
- **The aim:** find the best production decision that is **at least 95% feasible** for all possible realizations of the random variables $h_1(\xi)$ and $h_2(\xi)$



$$\begin{aligned} \min \quad & 2x_1 + 3x_2 \\ \text{s.t.} \quad & x_1 + x_2 \leq 100 \\ & \mathbb{P} \left[\begin{array}{l} 2x_1 + 6x_2 \geq h_1(\xi) \\ 3x_1 + 3x_2 \geq h_2(\xi) \end{array} \right] \geq 0.95 \\ & x_1, x_2 \geq 0 \end{aligned}$$

- Well defined model
- Tractable due to the assumption of the probability distribution of $h_1(\xi)$ and $h_2(\xi)$, and approximation:

$$\theta_1 \sim \mathcal{N}(0, 12) \quad \rightarrow \quad h_1(\xi) = 180 + \theta_1 \sim \mathcal{N}(180, 12)$$

$$\theta_2 \sim \mathcal{N}(0, 9) \quad \rightarrow \quad h_2(\xi) = 162 + \theta_2 \sim \mathcal{N}(162, 9)$$



- The resulting approximated deterministic LP:

$$\min \quad 2x_1 + 3x_2$$

$$\text{s.t.} \quad x_1 + x_2 \leq 100$$

$$2x_1 + 6x_2 \geq F_{h_1}^{-1}(0.975) = 180 + \Phi^{-1}(0.975) \cdot 12$$

$$3x_1 + 3x_2 \geq F_{h_2}^{-1}(0.975) = 162 + \Phi^{-1}(0.975) \cdot 9$$

$$x_1, x_2 \geq 0$$



- Compared to the corresponding models*

Model	x_1	x_2	Production cost	Feasibility
Deterministic (EV)	36	18	126	25%
Worst-case (RO)	48.02	25.55	172.681	100%
Recourse model	37.94	21.04	138.59	91%
Chance-constrained	38.94	20.94	140.70	95%

- Increased cost and feasibility
- More flexible/direct to control the feasibility than the recourse model

* Without productivity uncertainty



- Probabilistic constraints (chance-constrained) models

$$\min_{x \in \mathbb{R}^n} \mathbb{E}[f_0(x, \xi)]$$

$$\text{s.t. } \mathbb{P}[f_i(x, \xi) \leq 0, \forall i = 1 \dots m] \geq 1 - \alpha$$

- In general, **very challenging** problems:
 - Computing numerically the probability is only possible for a few classes of distributions
 - Feasible set is generally **nonconvex**: discrete cases resulting in MILP
 - Harder than recourse models
 - Convex reformulations (specific distributions) and Monte Carlo sampling approach (general)



Simple example of nonconvexity

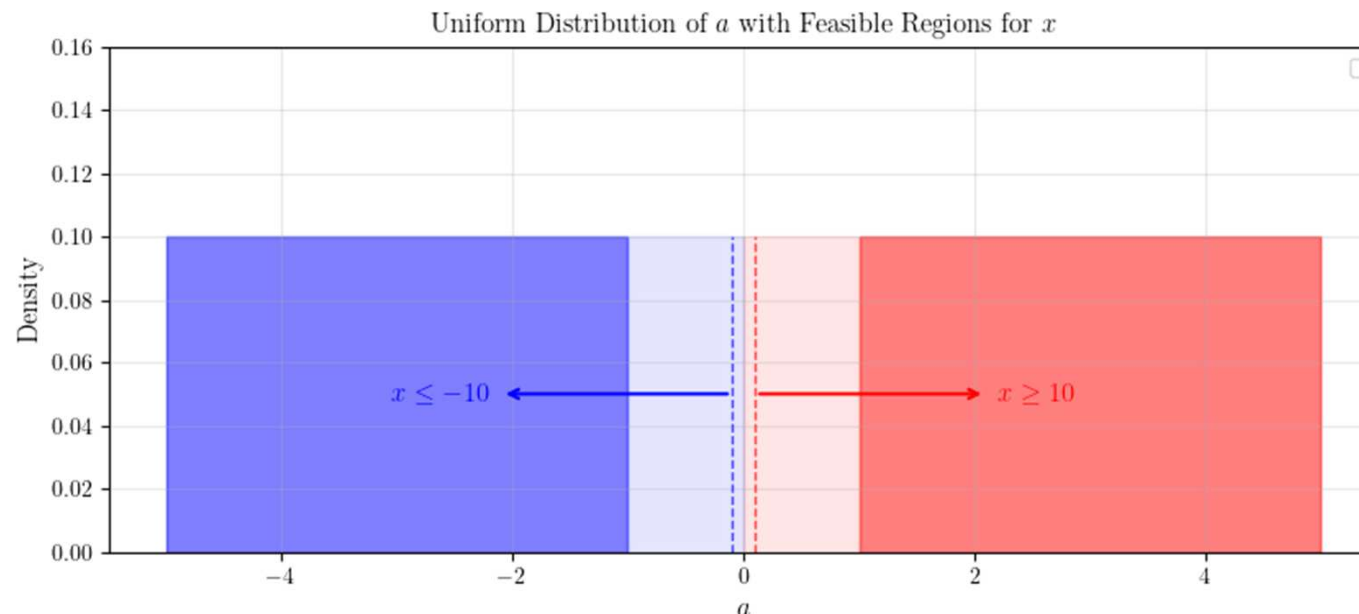


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$$\mathbb{P}(ax \geq 10) \geq 0.4, \text{ where } a \sim \text{Uniform}[-5, 5]$$

$$\begin{cases} \text{if } x > 0, \mathbb{P}(a \geq \frac{10}{x}) \geq 0.4 & \Rightarrow \frac{10}{x} \leq 1 \Rightarrow x \geq 10 \\ \text{if } x < 0, \mathbb{P}(a \leq \frac{10}{x}) \geq 0.4 & \Rightarrow \frac{10}{x} \geq -1 \Rightarrow x \leq -10 \\ x = 0 \text{ is infeasible} \end{cases}$$

$$\Rightarrow x \in (-\infty, -10] \cup [10, +\infty)$$





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$$\min_{x \in \mathbb{R}^d} \mathbb{E}[f_0(x, \xi)]$$

$$\text{s.t. } \mathbb{P}[f_i(x, \xi) \leq 0, \forall i = 1 \dots m] \geq 1 - \alpha$$

- Assume $\xi \in \{\xi_1, \xi_2, \dots, \xi_N\}$ where ξ_n has probability p_n , define

$$\mathbb{I}([f_1, f_2, \dots, f_m]) = \begin{cases} 0 & \text{if } f_i \leq 0, \forall i \in \{1, \dots, m\} \\ 1 & \text{if } f_i > 0, \exists i \in \{1, \dots, m\} \end{cases}$$

- The chance constraint is **equivalent** to

$$\sum_{n=1}^N p_n \mathbb{I}([f_1(x, \xi_n), f_2(x, \xi_n), \dots, f_m(x, \xi_n)]) \leq \alpha$$

- The chance constraint **is equivalent to**

$$\sum_{n=1}^N p_n \mathbb{I}([f_1(x, \xi_n), f_2(x, \xi_n), \dots, f_m(x, \xi_n)]) \leq \alpha$$

- Using a **binary variable** z_n to play the indicator function's role

$$\Leftrightarrow \begin{cases} \sum_{n=1}^N p_n z_n \leq \alpha \\ f_i(x, \xi_n) \leq z_n \text{UB}_i, \quad i = 1, \dots, m, \quad n = 1, \dots, N \\ z_n \in \{0, 1\}, \quad n = 1, \dots, N \end{cases}$$

- An (exact) mixed integer programming (MILP) formulation
- Non-convex, how to solve?



$$\min \quad 2x_1 + 3x_2$$

$$\text{s.t.} \quad x_1 + x_2 \leq 100$$

$$\mathbb{P} \left[\begin{array}{l} (2 + \eta_1)x_1 + 6x_2 \geq 180 + \theta_1 \\ 3x_1 + (3.4 - \eta_2)x_2 \geq 162 + \theta_2 \end{array} \right] \geq 1 - 0.05$$

$$x_1, x_2 \geq 0$$

- Suppose scenarios $[\eta_1^n, \eta_2^n, \theta_1^n, \theta_2^n]$, $n = 1, \dots, N$, each with p_n

$$\left\{ \begin{array}{l} \sum_{n=1}^N z_n \cdot p_n \leq 0.05 \\ 180 + \theta_1^n - [(2 + \eta_1^n)x_1 + 6x_2] \leq z_n \text{UB}_1, \quad n = 1, \dots, N \\ 162 + \theta_2^n - [3x_1 + (3.4 - \eta_2^n)x_2] \leq z_n \text{UB}_2, \quad n = 1, \dots, N \\ z_n \in \{0, 1\}, \quad n = 1, \dots, N \end{array} \right.$$

- **Challenges:** for large values of N , the MILP may become very difficult to solve
- **Strategies:**
 - tighter formulations (valid inequalities)
 - decomposition
 - parallel computing
 - integrating machine learning (e.g., learn to branching, very active research direction)
 - early stop in practice



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- **Log-concave distributions** if

$\log f_{\xi}(z)$ is concave over the whole support

→ $e^{-g(z)}$ for convex g is log-concave

→ Normal, Exponential, Uniform, Logistic, etc.

- The **CDF** of a log-concave distribution is also log-concave
- **Unimodality**: Log-concave distributions are unimodal, simplifying quantile-based analysis.



$$\mathbb{P}(a_i^\top x + b_i \leq \xi_i, \forall i = 1 \dots m) \geq 1 - \alpha$$

- results in **convex feasible set** for
 - 1) **fixed** a_i , and
 - 2) r.v. ξ_i **independent** and
 - 3) each follows **log-concave distribution**

$$\Rightarrow \prod_{i=1}^m \mathbb{P}(a_i^\top x + b_i \leq \xi_i) \geq 1 - \alpha$$

$$\Rightarrow \sum_{i=1}^m \log \mathbb{P}(a_i^\top x + b_i \leq \xi_i) \geq \log(1 - \alpha)$$

- Upper-level set of a concave function, thus convex



Example



$$\mathbb{P}(a_i^\top x + b_i \leq \xi_i, \forall i = 1 \dots m) \geq 1 - \alpha \text{ where } \xi_i \sim \text{Exp}(\lambda_i)$$

$$\Rightarrow \mathbb{P}(a_i^\top x + b_i \leq \xi_i, \forall i = 1 \dots m)$$

$$= \prod_{i=1}^m e^{-\lambda_i (a_i^\top x + b_i)}$$

$$= e^{-\sum_{i=1}^m \lambda_i (a_i^\top x + b_i)} \geq 1 - \alpha$$

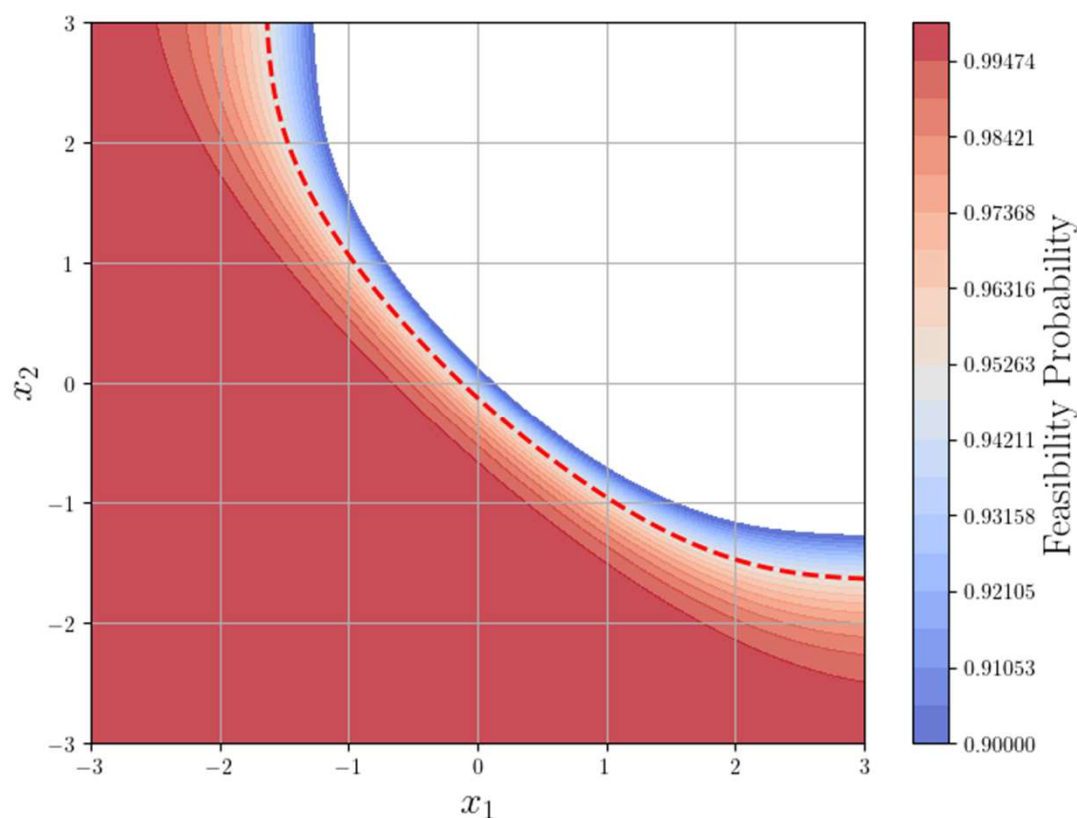
$$\Rightarrow \sum_{i=1}^m \lambda_i (a_i^\top x + b_i) \leq -\log(1 - \alpha)$$

- Feasible set is a **polyhedron**, thus convex



- When having **dependent uncertainties** among constraints
- Not convex even with multivariate normal (Log-concave), and only RHS uncertainty

$$\mathbb{P}(x_1 \leq \xi_1, x_2 \leq \xi_2) \geq 0.95 \quad (\xi_1, \xi_2) \sim \mathcal{N} \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 & -0.9 \\ -0.9 & 1 \end{bmatrix} \right)$$





- For **dependent uncertainties**, joint chance constraints

$$\mathbb{P} \left(\cap_{i=1}^m \mathcal{C}_i \right) = 1 - \mathbb{P} \left(\cup_{i=1}^m \bar{\mathcal{C}}_i \right) \geq 1 - \alpha$$

- Allowable violation probability

$$\Leftrightarrow \mathbb{P} \left(\cup_{i=1}^m \bar{\mathcal{C}}_i \right) \leq \alpha$$

- Union bound (Boole's inequality)

$$\mathbb{P} \left(\cup_{i=1}^m \bar{\mathcal{C}}_i \right) \leq \sum_{i=1}^m \mathbb{P} \left(\bar{\mathcal{C}}_i \right) \leq \alpha \stackrel{\text{cons. approx.}}{\implies} \mathbb{P} \left(\bar{\mathcal{C}}_i \right) \leq \frac{\alpha}{m}$$

- Decompose into m independent individual constraints

$$\mathbb{P} \left(\mathcal{C}_i \right) \geq 1 - \frac{\alpha}{m}, \quad \forall i = 1 \dots m$$



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- **Scenario approximation**, inspired by RO
- Distribution-free, able to handle general dependencies among constraints
- **Idea**: satisfy constraints for **a set of samples** and then the solution satisfies for an overall probability with some confidence

$$\mathbb{P}[f_{[1\dots m]}(x, \xi) \leq 0] \geq 1 - \alpha$$

- Use i.i.d samples $\xi_1, \xi_2, \dots, \xi_N$ to solve

$$f_{[1\dots m]}(x, \xi_i) \leq 0, \quad \forall i = 1, \dots, N \quad \Rightarrow \quad \hat{x}_N$$

- **Simpler** than the MIP formulation which uses binary variables



- What sample size N is needed to make sure that $\hat{x}_N$ is feasible to the original problem?

$$\mathbb{P}[f_{[1\dots m]}(\hat{x}_N, \xi) \leq 0] \geq 1 - \alpha \quad (*)$$

- Cannot ensure...instead with high confidence $1 - \beta$

$$\mathbb{P}(\hat{x}_N \text{ violate } *) \leq \sum_{k=0}^{d-1} \binom{N}{k} \alpha^k (1 - \alpha)^{N-k} \leq \beta \quad (**)$$

- Simplified to (by Chernoff inequality):

$$N \geq \frac{d - 1 + \overbrace{C}^{C=\ln(\beta^{-1})} + \sqrt{C[2(d-1) + C]}}{\alpha}$$

Calafiore, G. C., & Campi, M. C. (2006). The scenario approach to robust control design. *IEEE Transactions on automatic control*, 51(5), 742-753.

Campi, M. C., & Garatti, S. (2008). The exact feasibility of randomized solutions of uncertain convex programs. *SIAM Journal on Optimization*, 19(3), 1211-1230.



- Sample size requirement

$$N \geq \frac{d - 1 + C + \sqrt{C[2(d - 1) + C]}}{\alpha}$$

- $N \geq 13410$ for $d = 100, \alpha = \beta = 0.01$
- $N \geq 12460$ if using the binomial bound (**)
- Might be too conservative in practice



A simple problem

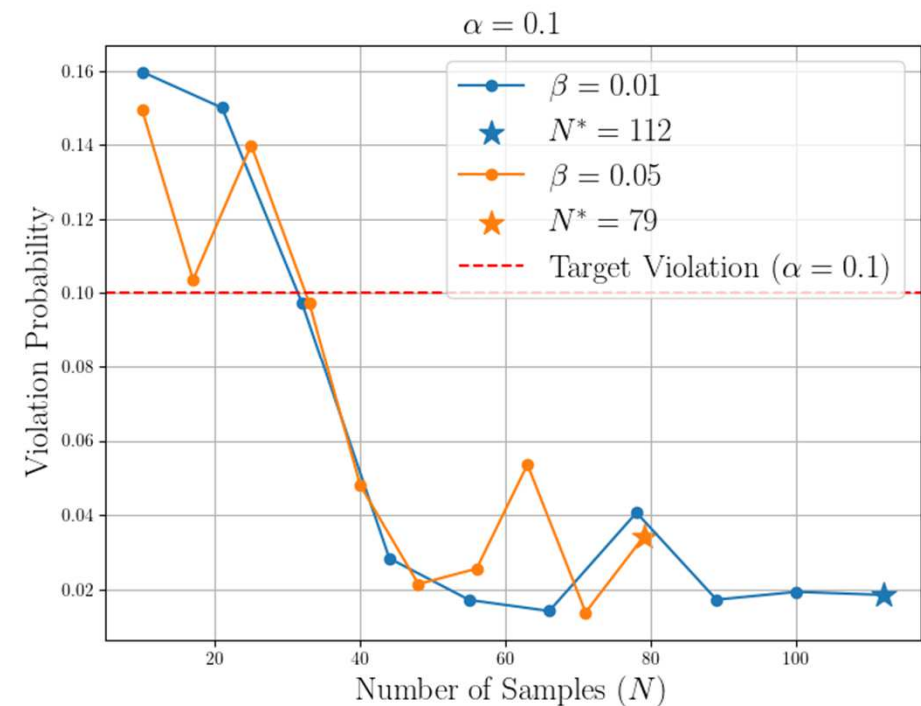
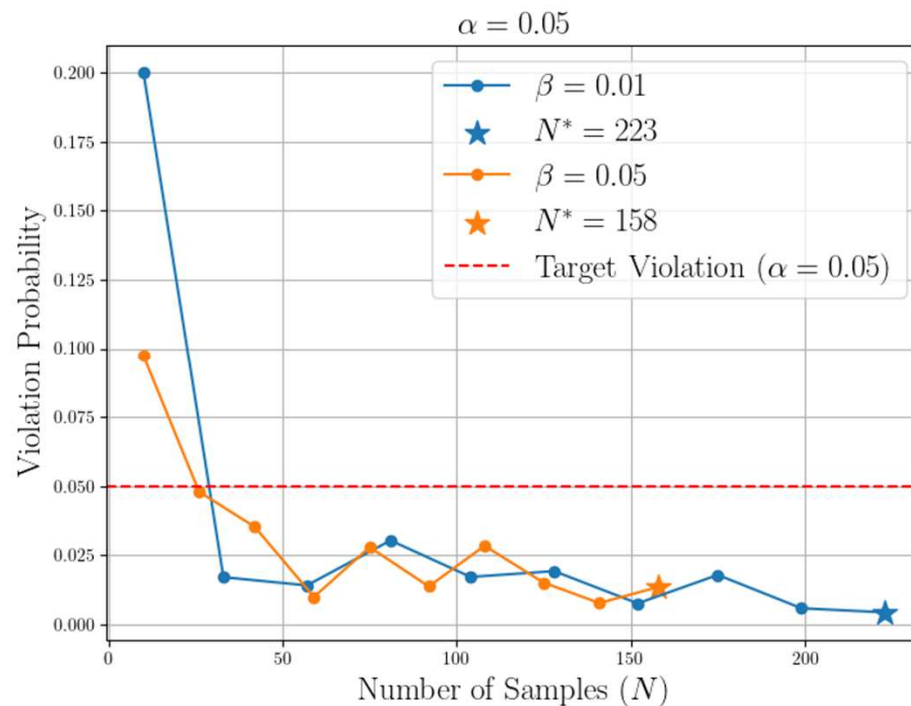


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$$\max_{x_1, x_2} x_1 + x_2$$

$$\text{s.t. } \mathbb{P}(\xi_1 x_1 + \xi_2 x_2 \leq 1) \geq 1 - \alpha \quad \text{where } (\xi_1, \xi_2) \sim \mathcal{N}(0, I)$$

$$x_1, x_2 \geq 0$$





- **Empirical calibration** based on **out-of-sample validation**
 - Start with a **smaller N** (e.g., 50% of the theoretic bound), solve the problem
 - Test solution feasibility on a **large independent validation data set** (e.g., of size M)
 - If the **empirical violation probability** is much lower than α , reduce N
 - Otherwise, increase N

$$\mathbb{P}_M[f_{[\exists 1 \dots m]}(\hat{x}_N, \xi) > 0] = \frac{1}{M} \sum_{j=1}^M \mathbb{I}(f_{[\exists 1 \dots m]}(\hat{x}_N, \xi_j) > 0)$$



- How to **evaluate the quality** of the solution from the scenario approximation?

- **Feasibility:**

construct CI and UB for the **empirical violation probability** (e.g., using resampling) on the validation set

$$\mathbb{P}_M^{\text{upper}}[f_{[\exists 1 \dots m]}(\hat{x}_N, \xi) > 0] \leq \alpha$$

- **Optimality**

the same as in SAA when the objective involves randomness



Example



$$\max_{x_1, x_2} x_1 + x_2$$

$$\text{s.t. } \mathbb{P}(\xi_1 x_1 + \xi_2 x_2 \leq 1) \geq 1 - \alpha \text{ where } (\xi_1, \xi_2) \sim \mathcal{N}(0, I)$$

$$x_1, x_2 \geq 0$$

- $\alpha = \beta = 0.05$
- Start from $N = 0.5N^* = 79$

$$[\hat{x}_1^{79} \ \hat{x}_2^{79}] = [0.4507 \ 0.3358] \text{ with obj} = 0.7865$$

$$\mathbb{P}_{10^5}^{\text{upper}}[\hat{x}_1^{79} \xi_1 + \hat{x}_2^{79} \xi_2 > 1] \approx 0.0387 < \alpha$$

- May decrease N to 60

$$[\hat{x}_1^{60} \ \hat{x}_2^{60}] = [0.4412 \ 0.3952] \text{ with obj} = 0.8362$$

$$\mathbb{P}_{10^5}^{\text{upper}}[\hat{x}_1^{60} \xi_1 + \hat{x}_2^{60} \xi_2 > 1] \approx 0.0481 \approx \alpha$$



Example



- If decrease N to 50

$$[\hat{x}_1^{50} \hat{x}_2^{50}] = [0.3960 \ 0.4882] \text{ with obj} = 0.8842$$

$$\mathbb{P}_{10^5}^{\text{upper}}[\hat{x}_1^{50} \xi_1 + \hat{x}_2^{50} \xi_2 > 1] \approx 0.0571 > \alpha$$

- Cannot do better than this, $N = 60$ is likely a good approximation

- True optimal:

$$[x_1^* \ x_2^*] = [0.4299 \ 0.4299] \text{ with obj} = 0.8596$$



- Chance-constrained models have **more flexibility** to control the solution feasibility than the recourse models
- But **computationally hard**, due to nonconvexity
- Convex reformulations are possible for **specific distributions** (log-concave) and constraint functions (fixed A)
- **Monte Carlo sampling (scenario approximation)** is distribution-free and can handle general dependencies among constraints
- Convex approximation/bound is also a widely used strategy (e.g., CVaR)



- Scalable algorithms for **high-dimensional** problems with **strongly correlated** uncertainties and **nonconvex** chance constraints (many **safe ML** applications)
- How to make sure **decision robustness** (out-of-sample performance)?
 - no exact probability distribution, but only **a few (not too many) data points**, common challenge for stochastic optimization
 - evolving distributions instead of static ones
 - e.g. DRO (Chapter 7 in Shapiro et al. 2021)



- Implement the chance constraint model for the gasoline production planning problem
- Use MILP and scenario approximation to solve the model



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